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RESEARCH-ARTICLE

Enhanced Image Segmentation for Indoor Autonomous Systems Using RealSense Camera and U-Net with Advanced Encoder and Attention Mechanism

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Enhanced Image Segmentation for Indoor Autonomous Systems Using RealSense Camera and U-Net with Advanced Encoder and Attention Mechanism

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Abstract

This paper introduces an advanced image segmentation approach utilizing the U-Net architecture with EfficientNetB3 as the encoder to enhance feature extraction in complex indoor environments. To further refine the segmentation process, an Attention-Gated Spatial Channel Squeeze Excitation (AG-scSE) mechanism is integrated, allowing the model to focus on key regions and optimize edge detection. The proposed method is evaluated using the NYUv2 dataset, segmenting up to 40 different classes to validate its effectiveness. Additionally, the study compares the performance of different EfficientNet encoders, ranging from B0 to B3, to identify the optimal configuration for segmentation tasks. Leveraging a RealSense camera as the vision system, this approach significantly improves the segmentation accuracy and spatial understanding of indoor autonomous systems by processing richer, more detailed data from the surrounding environment.

CCS Concepts

• **Applied computing**; • **Computer systems organization** → **Robotics**; • **Computing methodologies** → **Machine learning**; **Object detection**; **Image segmentation**;

Keywords

Indoor navigation, RealSense Camera, Object recognition, Semantic segmentation, EfficientNet, U-Net, NYUv2 dataset, Real-time obstacle detection, Home automation, Autonomous robot systems

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1 Introduction

In the rapidly advancing field of artificial intelligence, particularly in computer vision, transformative developments are redefining machine capabilities, making them more powerful and accurate [1, 2]. Image segmentation plays a pivotal role in this field, enabling the separation of image components for improved analysis and detection [4]. By utilizing image segmentation, machines can perceive images comprehensively, similar to human vision, allowing for better adaptation to diverse and dynamic environments.

The operation of Indoor Autonomous Vision Systems (IAVs) relies heavily on computer vision, especially image segmentation techniques, to achieve optimal efficiency in addressing specific challenges. These challenges include accurately identifying objects in cluttered and confined indoor spaces, handling complex lighting conditions, and distinguishing between overlapping or closely spaced objects. Traditional image processing methods, such as manual feature extraction, often fail to meet the demands of such intricate environments, leading to suboptimal performance [5–7]. The development of convolutional neural networks (CNNs) has proven essential in optimizing segmentation tasks and providing higher accuracy. Moreover, integrating RGB-D depth data provides crucial spatial and contextual information, improving the delineation of object boundaries and facilitating precise separation of objects based on their spatial positions [8–13].

To address these challenges, this study introduces the Improved EfficientNet-based U-Net (IEUnet), an enhanced version of the U-Net architecture. IEUnet incorporates modified parameters and additional attention mechanisms to improve segmentation accuracy. It employs EfficientNet as the encoder, comparing versions from B0 to B3 to enhance feature extraction. The primary goal is to determine the optimal configuration for segmentation, enabling the model to increase its capability from 13 to 40 classes while maintaining or improving accuracy. The contributions of this study include:

- Developing the Improved EfficientNet-based U-Net (IEUnet) by integrating EfficientNet encoders ranging from B0 to B3 and evaluating its performance on the NYUv2 dataset with 40 classes.
- Deploying the IEUnet model in a RealSense camera system to assess its real-time segmentation performance, enabling

accurate detection in indoor environments with enhanced object recognition capabilities.

These contributions are significant for advancing indoor computer vision systems by addressing critical gaps in segmentation accuracy and efficiency. Experimental results demonstrate that IEUnet achieves 32% mIoU on the validation set, surpassing baseline U-Net models in handling the complex indoor scenes of the NYUv2 dataset. Furthermore, its deployment in the RealSense camera system confirms its capability for real-time segmentation, ensuring robust detection and classification across various indoor environments. These findings highlight the model’s potential for practical applications, such as autonomous robotics and smart home systems.

2 Related Work

In image processing, convolutional neural networks (CNNs) have greatly advanced the accuracy of image segmentation, especially with architectures such as U-Net. Originally developed by Ronneberger et al. (2015) [14], U-Net has become widely recognized in medical image segmentation and other areas due to its ability to capture and retain intricate features from input data. Its unique encoder-decoder structure with skip connections effectively combines low-level details with high-level contextual information, resulting in notable improvements in segmentation accuracy. This success has inspired adaptations of U-Net across various domains, extending its application to tasks beyond medical imaging, such as indoor scene segmentation with RGB-D data.

However, many of these tasks, particularly in medical imaging and scene segmentation, often encounter imbalanced scenarios where certain classes (e.g., rare diseases or underrepresented objects) have significantly fewer samples than others [21, 22]. This imbalance poses challenges in model training and can lead to biased predictions favoring majority classes. Imbalanced learning has become a crucial focus in segmentation tasks, with strategies such as class weighting, synthetic data generation, and architecture optimization being explored to address these challenges, Cheng and Mai et al. (2023) [20].

Additionally, incorporating pre-trained networks—especially EfficientNet—into U-Net models has further enhanced feature extraction for complex data. Tan and Le (2019) [3] introduced EfficientNet, a family of CNNs optimized by balancing depth, width, and resolution, achieving superior accuracy while minimizing computational costs. Integrating EfficientNet into U-Net frameworks has improved performance on high-resolution and complex images. In recent work, Thu et al. [16] applied their modified U-Net, EADFL-UNet, to the NYUv2 13-class dataset, achieving a 10% improvement in mIoU over conventional architectures, even without utilizing depth information.

Other researchers, such as Badrinarayanan et al. (2017) with SegNet [17], have underscored the importance of encoder-decoder structures for segmentation tasks. Similarly, Chen et al. (2018) with DeepLab [18] have leveraged atrous spatial pyramid pooling to capture multi-scale information, adding to the pool of effective segmentation models.

Given the progress in segmentation accuracy achieved by architectures, our study proposes IEUnet, an enhanced U-Net architecture with EfficientNetB0–B3 encoders to improve segmentation

precision for up to 40 classes from the NYUv2 dataset. By combining the strengths of U-Net and EfficientNet, alongside AG-scSE attention blocks, IEUnet aims to advance the capabilities of RGB-D segmentation for indoor environments.

3 Dataset

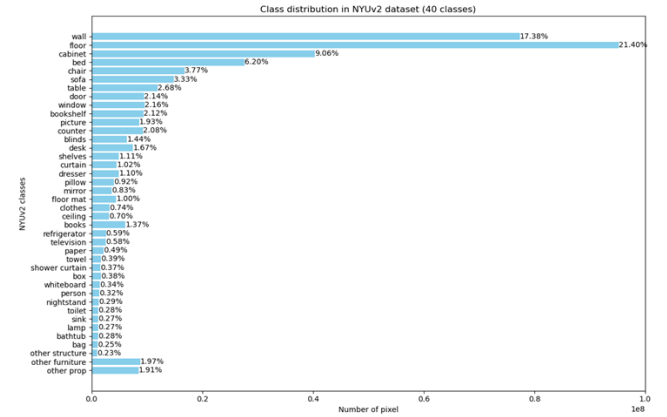


Figure 1: Class distribution in NYUv2 dataset

The NYUv2 dataset [18] is a well-established and widely recognized dataset in the field of computer vision, specifically designed for semantic segmentation tasks in indoor environments. This dataset is constructed from 1,449 video sequences captured across various indoor settings, including living rooms, kitchens, and offices, using a Kinect sensor. From these videos, researchers have extracted 407,024 frames, including 1,449 fully annotated RGB-D images, providing pixel-by-pixel labels with detailed information.

The most utilized version of NYUv2 features 40 segmentation classes (Figure 1), making it highly valuable for training and testing models. It includes 795 samples for training and 654 samples for testing, with object classes covering common indoor elements such as floors, walls, ceilings, windows, chairs, tables, beds, doors, cabinets, lamps, people, books, screens, pictures, trash cans, refrigerators, sinks, among others. This diversity adds to the complexity of object recognition and segmentation tasks, making it suitable for dynamic and varied indoor environments.

Images in the NYUv2 dataset have a resolution of 640x480 pixels, offering sufficient detail for analyzing object features. Notably, NYUv2 provides RGB and depth data, significantly enhancing 3D segmentation and recognition models. With its extensive number of images, high annotation accuracy, and detailed labels, NYUv2 is a benchmark dataset for evaluating computer vision models, particularly in autonomous robot applications and home automation systems.

4 Methodology

This study presents an IEUnet framework, aiming to expand the segmentation capability to include a larger number of object classes in indoor environments, specifically targeting 40 classes using the NYUv2 dataset. The methodology comprises three core components: the design of IEUnet architecture, the training procedure for optimal

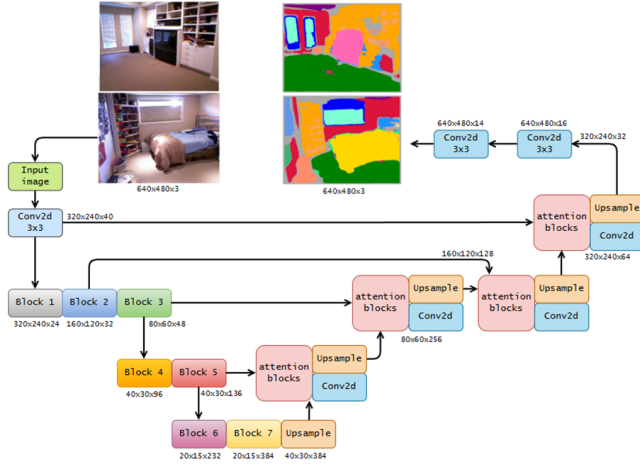


Figure 2: Model architecture

performance, and the real-time evaluation setup using a RealSense camera.

4.1 IEUNET Architecture

The architecture of IEUnet is illustrated in Figure 2, which shows the integration of EfficientNet as the encoder-backbone and highlights the use of Attention Blocks as well as the upsampling and convolutional blocks designed to enhance segmentation accuracy across multiple object classes. This figure underscores the structured flow from input to output, demonstrating how each component integrates. Main components that constructed the IEUnet are:

- **Encoder-Backbone Integration:** To optimize feature extraction from complex indoor scenes, IEUnet uses blocks 1 to 7 (Figure 2) as encoders. These blocks implement multiple versions of EfficientNet (B0 to B3), each providing a different balance between depth and performance. Comparisons are performed to determine which encoder can achieve the best segmentation performance when faced with increasing complexity of object classes. The EfficientNet encoders are pre-trained, leveraging the powerful feature extraction capabilities to enrich subsequent segmentation layers.
- **Attention Mechanism:** IEUnet architecture incorporates Attention Blocks, which include a combination of Attention-Gated Spatial Channel Squeeze Excitation (AG-scSE) mechanisms. These attention mechanisms enable the model to focus on essential spatial and channel-wise features, improving its ability to accurately segment complex objects. This Attention mechanism has been mentioned in the paper of Thu et al.[15], demonstrating its ability to enhance the efficiency of scene segmentation and support indoor automatic vision systems.
- **Integration of Depth Information:** Using RGB-D images as input, the framework incorporates depth data alongside RGB channels. The depth of information enhances the model's ability to understand object boundaries and spatial relationships, which is particularly valuable in indoor environments with closely spaced objects. This helps distinguish

between objects based on their relative depth, a crucial factor for fine-grained segmentation across 40 object classes.

4.2 Training Process

The IEUnet framework is trained on the NYUv2 dataset, which contains annotations for 40 distinct object classes frequently found in indoor environments. The training process is carefully crafted to address the increased complexity of handling more classes:

- **Data Preprocessing:** Training data consisted of segmented images and ground-truth masks. Each image-mask pair was loaded and batched via DataLoader, which randomly sampled patches to enrich the dataset and aid in generalization. Input data was processed to standard dimensions and normalized before being fed into the model.
- **Model Setup:** A deep learning segmentation model was trained with a modified UNet architecture utilizing EfficientNet-b3 as the backbone for feature extraction. The model was adapted for pixel-level classification to handle a multi-class segmentation task on the NYUv2 dataset.
- **Loss Function:** A custom Focal Loss with Dice Loss (Class-Balanced Weights Focalloss or CBW-FL) [15] integration was employed to manage class imbalance and enhance segmentation precision. The Focal Loss focused on difficult-to-classify pixels, while the Dice Loss ensured balanced region segmentation. CBW-FL is defined as:

$$L_{CBW-FL} = -\frac{1}{N} \sum_{i=1}^N \sum_{j=1}^m \omega_i (1 - y_{ij}^{\text{pred}}) y_{ij}^{\text{true}} \log(y_{ij}^{\text{pred}}) \quad (1)$$

where ω_c is the class weight:

$$\omega_c = 1 - \frac{f_c}{\sum_{i=1}^C f_i} \quad (2)$$

- N : Total number of samples.
- m : Number of class.
- y_{ij}^{pred} : Predicted probability for the j -th class in the i -th sample.
- y_{ij}^{true} : Ground-truth label for the j -th class in the i -th sample.
- f_c : Frequency of the c -th class.
- C : Total number of class.
- **Hyperparameters:** To prevent overfitting, weight decay (10^{-4}) was applied. The training spanned 20 epochs, with early stopping based on validation performance.

4.3 Evaluation on RealSense Camera

To evaluate the model in a practical setting, the IEUnet framework is deployed using a RealSense camera in an indoor environment to simulate real-time performance:

Real-Time Segmentation. The model is connected to a RealSense camera setup that streams RGB-D data in real-time. Segmentation predictions for 40 object classes are processed live, allowing observation of the framework's responsiveness and adaptability.

Performance Metrics. Key metrics such as mean Intersection over Union (mIoU) and segmentation accuracy are recorded, focusing on the model's ability to identify and differentiate between a broad range of indoor objects. Performance across EfficientNet

encoder versions is compared to assess real-time accuracy and computational efficiency.

Qualitative Analysis. Sample segmentation outputs are visually analyzed to ensure the model maintains object boundaries, accurately labels each segmented object, and effectively responds to real-time inputs.

By designing the IEUnet framework with expanded segmentation capabilities and incorporating EfficientNet-based encoders, this methodology supports a robust, high-performance solution for IAVs. The integration of SABs and multi-class segmentation provides a scalable approach for real-time applications in autonomous robotics and indoor automation systems.

5 Experiment

In this experiment, we evaluated U-Net with different EfficientNet backbones (B0, B1, B2, and B3) for a 40-class segmentation task using the NYUv2 dataset. We tracked the training and validation performance of each model in terms of loss, mean Intersection over Union (mIoU), and pixel accuracy.

5.1 Experimental Setup

The experiments were conducted on a Windows 10 Pro system with a 12th Gen Intel® Core™ i5-12400F processor, featuring 6 cores and 12 logical processors and a base frequency of 2500 MHz. The system is equipped with 16 GB of installed physical RAM, offering 15.8 GB of total physical memory and 41.8 GB of total virtual memory. The motherboard used is an ASUS PRIME B660M-A D4, Rev 1. xx, with an American Megatrends Inc. BIOS ver1620, operating in UEFI mode. Additionally, the system includes an NVIDIA GeForce RTX 2060 graphics card, providing ample computational power for deep learning tasks. On the software side, the experiments were implemented using Jupyter Notebook, with Anaconda serving as the environment and package manager. The primary programming language used was Python, with the EfficientNet model employed for semantic segmentation tasks. This setup proved to be efficient, allowing all 4 models to run effectively at the same time on this configuration.

5.2 Evaluation Metrics

Firstly, pixel accuracy measures the percentage of pixels that are classified correctly across the entire output. To achieve this, the output is processed through a softmax function to obtain probabilities, followed by an argmax operation to select the class with the highest probability. Each pixel is then compared with its ground-truth class; if they match, that pixel is considered accurate. Pixel accuracy is calculated as the ratio of correctly classified pixels to the total number of pixels, with the formula:

$$\text{Pix_acc} = \sum_i \frac{n_{ii}}{t_i} \quad (3)$$

Secondly, the mIoU metric is a widely used evaluation method for segmentation that averages the Intersection over Union (IoU) for each class. For each class i , mIoU calculates the intersection and union between pixels in the predicted mask and the ground-truth mask for that class. The intersection, n_{ii} , represents the count of pixels where both the predicted and ground-truth masks indicate

the class i presence, while the union is calculated as $t_i + \sum_{j \neq i} n_{ji} - n_{ii}$, where t_i is the total number of ground-truth pixels in class i , and n_{ji} is the count of pixels predicted as class j but belonging to class i .

The mIoU is then calculated by averaging the IoU for each class across all classes, as follows:

$$\text{mIoU} = \frac{1}{n_{\text{class}}} \sum_{i=1}^{n_{\text{class}}} \frac{n_{ii}}{t_i + \sum_{j \neq i} n_{ji} - n_{ii}} \quad (4)$$

where n_{class} is the total number of classes. This metric avoids division by zero by excluding classes with no ground-truth instances. The two metrics, Pixel Accuracy and mIoU, are computed at each epoch and used to track the model's performance on both the training and validation datasets.

5.3 Experimental Results

Model	Train loss	Val loss	Train mIoU	Val mIoU	Train acc	Val acc
E-B0	0.586	0.651	0.327	0.301	0.616	0.587
E-B1	0.585	0.628	0.325	0.300	0.618	0.594
E-B2	0.562	0.562	0.338	0.311	0.630	0.599
E-B3	0.538	0.599	0.355	0.322	0.641	0.614

Table 1: Comparison between encoders as EfficientNet B0 - B3

Based on the experimental results in Table 1 for U-Net with various EfficientNet backbones (B0 to B3) on a 40-class segmentation task, we observed a steady improvement in performance with deeper backbones, with EfficientNet-B3 yielding the best results among all configurations.

The U-Net model with EfficientNet-B0 as the backbone showed relatively high error rates, with a training loss of 0.586 and a validation loss of 0.651. This configuration resulted in a training mIoU of 0.327 and validation mIoU of 0.301, alongside accuracy scores of 0.616 for training and 0.587 for validation. These values suggest that the model struggled to accurately segment multiple classes, likely due to the limited capacity of the B0 backbone.

Moving to EfficientNet-B1, we observed a slight decrease in validation loss to 0.628 and an increase in validation accuracy to 0.594, with training accuracy improving to 0.618. However, the more values remained similar to B0, showing only minor improvements. This indicates that the B1 backbone had a limited impact on enhancing the segmentation performance.

EfficientNet-B2 showed a further improvement in both loss reduction and accuracy, with a training loss of 0.562, validation loss of 0.562, and mIoU scores of 0.338 for training and 0.311 for validation. The accuracy scores reached 0.630 for training and 0.599 for validation, reflecting improved feature extraction capabilities and a stronger segmentation performance compared to the previous backbones.

Finally, the EfficientNet-B3 backbone achieved the best performance, with a training loss of 0.538 and a validation loss of 0.599. The training mIoU reached 0.355, while the validation mIoU was

0.322, slightly higher than the B2 configuration. Accuracy also increased, with 0.641 for training and 0.614 for validation. This model achieves the best performance largely due to its ability to balance depth, width, and resolution from the compound scaling mechanism, proving it to be the optimal choice among the backbones tested for this particular task.

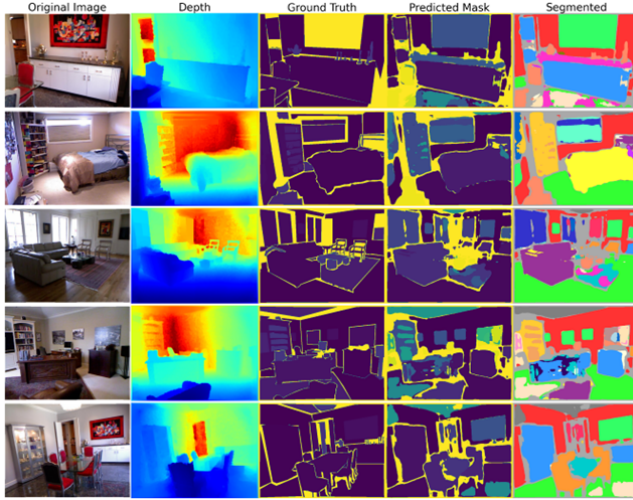


Figure 3: Segmented Results with EfficientNet-B3 as encoder

The segmented results when using EfficientNet-B3 in Figure 3 show the effective segmentation of images. The depth columns show the RGB-D images using a RealSense camera and the segmented columns are images that have been color-coded for each object recognized from the Original Image, general objects such as Walls, Floors, Cabinets, Doors or objects with many detailed angles such as Books, Chair, Sofar, Table.. are segmented very easily recognizable when looking at the results, different colors are used to distinguish one type of object from the others to clarify the separation of edges and edges of objects after segmentation.

5.4 Comparative Analysis

Previous works on semantic segmentation have primarily leveraged standard convolutional networks or the traditional U-Net architecture, which, although effective, often struggles with maintaining performance efficiency and robustness, particularly when handling complex indoor environments with numerous object classes.

This study introduces an enhanced and optimized U-Net architecture that incorporates the EfficientNet backbone and is fine-tuned specifically for indoor environments using the NYUv2 dataset. By leveraging these advanced techniques, the model achieves a significant balance between accuracy and efficiency. The proposed model achieves a notable mIoU of 32% on validation, making it particularly suited for real-time applications in autonomous home robots, where real-time processing and low-latency responses are critical.

Moreover, the comparative analysis shows that this study addresses the limitation of handling diverse indoor objects more effectively than previous research. By training with a larger set of

class-specific labels (40 classes in the NYUv2 dataset), the model demonstrates improved generalization capabilities and a more precise segmentation in intricate environments. Additionally, the integration with a depth-vision RealSense camera further enhances the accuracy of object localization and spatial understanding, setting it apart from previous studies that primarily rely on RGB cameras. This hybrid approach provides a more comprehensive spatial and depth-based understanding, which is particularly advantageous for robotic vision in indoor autonomous applications.

5.5 Discussion

The discussion focuses on several key points regarding the performance of the model and the potential for future improvements. First, the pixel accuracy of the IEUnet model remains stable even when the number of objects to be segmented increases significantly. This shows that the model is capable of effectively handling complex scenes with a larger number of objects, a significant improvement over previous studies in this area.

Although the experimental results are not really impressive, the accuracy achieved is still high enough to consider the feasibility of the model in practical applications, especially for segmentation problems that require similar accuracy levels.

In addition, an important factor that can help improve the performance of the model is the addition of depth information. Using depth data is capable of providing more detailed spatial information, assisting the model in better distinguishing between object classes and improving the quality of the segmentation results.

6 Conclusion

In this study, we proposed a method to improve image segmentation in indoor environments by developing the IEUnet architecture, integrating U-Net as the backbone, and utilizing EfficientNetB0 to B3 models as encoders. This approach not only extends segmentation capability to 40 object classes from the NYUv2 dataset but also enhances segmentation accuracy, particularly in identifying key boundaries and regions with the help of AG-scSE attention blocks.

Experimental results indicate that EfficientNet-B3 achieved the best performance among the tested models, with significant improvements in both accuracy and more. These findings suggest that incorporating depth information from RealSense cameras, combined with deep learning methods, can considerably advance the effectiveness of autonomous robotic systems in indoor environments.

The contributions of this research not only introduce a novel approach to image segmentation but also provide a foundation for further studies in home automation and object recognition. In the future, we aim to continue refining the model and expanding the practical applications of IEUnet to more complex scenarios, enhancing robots' ability to recognize and interact within human living spaces.

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